06-10-2025
Monday

KINETICS OF COMPLEX R^n's.

* complex R^n : R^n's with multiple steps.
* For such a R^n, the rate depends on the slowest step. eg; SN_1 R^n
* There are Four types of complex R^n's.

① opposing / Reversible.

② consecutive.

③ Parallel.

④ chain.

1.] opposing / Reversible R^n.

$$A \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} B$$

	A	B
$t=0$	a	0
t	$(a-x)$	x

$\frac{dx}{dt} = k_1(a-x)$ → Forward part only.

$\therefore \frac{dx}{dt} = k_1 \cdot (a-x) - k_{-1} \cdot x$

at equilibrium ; $\frac{dx}{dt} = 0$

$\therefore \boxed{k_1 \cdot (a-x) = k_{-1} \cdot x}$

2.] Consecutive R^n.

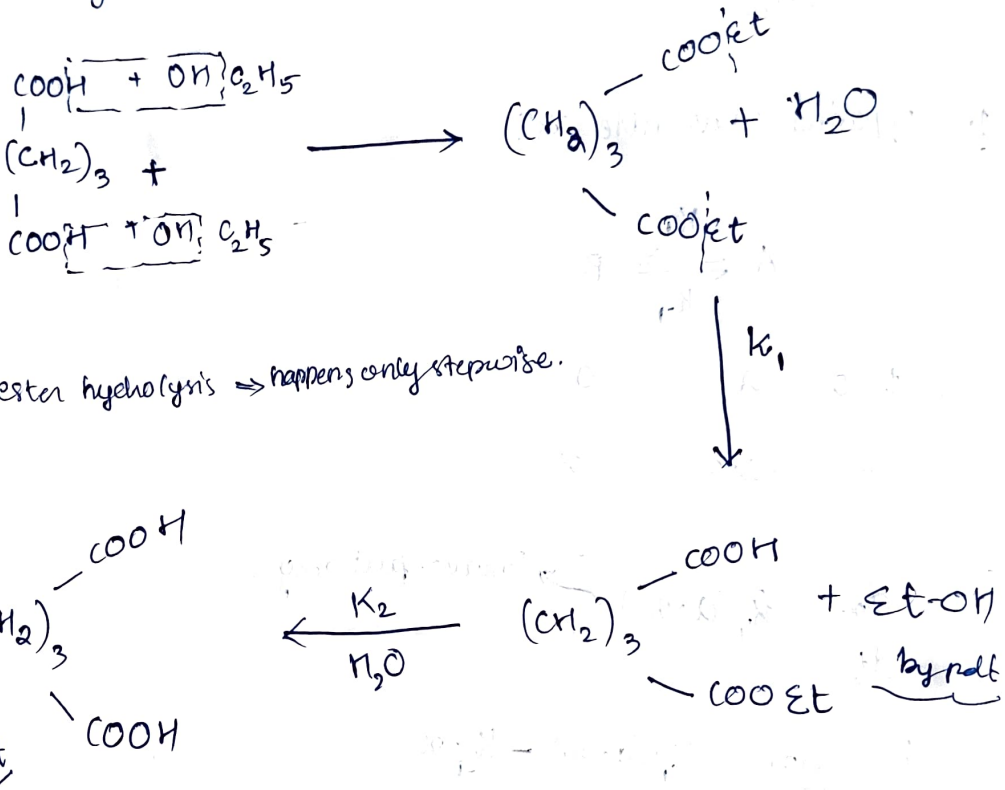
$$A \xrightarrow{k_1} B \xrightarrow{k_2} C$$

$$-\frac{d[A]}{dt} = k_1 \cdot x$$

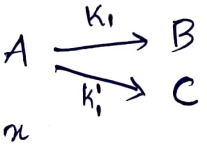
$$\frac{d[B]}{dt} = k_1 x - k_2 y$$

$$\frac{d[C]}{dt} = k_2 y$$

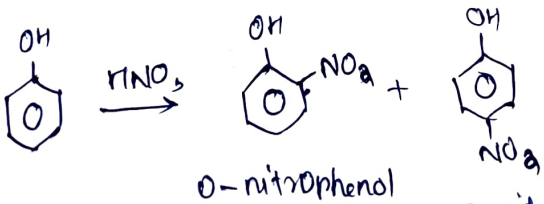
eg: Diethyl adipate.



3.] parallel R^n.



both pdt are important equimolar and equally formed.



o-nitrophenol

p-nitrophenol.

(almost equal quantity)

$$\frac{d[B]}{dt} = k_1 x$$

$$\frac{d[C]}{dt} = k_1' x$$

07-10-25
Tuesday

4.] Chain Rxn:

Chain rxns are initiated by free radicals.

1. * chain initiation.
2. * chain propagation. } same
3. * chain transfer
4. * chain termination. (radical combine each other to give neutral molecules)

$$H_2 + Cl_2 \xrightarrow{h\nu} 2HCl$$

1. $Cl-Cl \xrightarrow{h\nu} 2Cl\cdot$

2. $Cl\cdot + H-H \longrightarrow HCl + H\cdot$

3. $H\cdot + Cl-Cl \longrightarrow HCl + Cl\cdot$

4. $H\cdot + H\cdot \longrightarrow H_2$

   $Cl\cdot + Cl\cdot \longrightarrow Cl_2$

   $H\cdot + Cl\cdot \longrightarrow HCl$

⇒ **Pseudo order Rxn.**

The chemical reactions appears to be of higher order, but actually follows lower order kinetics.

eg: $CH_3COOC_2H_5 + H_2O \xrightarrow{H^+} CH_3COOH + C_2H_5OH$

Ethyl ethanoate (Ethyl acetate)

Ethanoic acid (Acetic acid) + Ethanol

Acid Catalysed Hydrolysis of Ester.

This rⁿ which is expected to be 2nd order is actually of first order. since water is excess, the order doesn't depend upon the con. of H_2O.

$$\text{rate} = K [CH_3COOC_2H_5] [H_2O]$$

$[H_2O]$ → constant

$$\text{rate} = K_1 \times K' [CH_3COOC_2H_5]$$

$$\text{rate} \propto [\text{ester}]$$

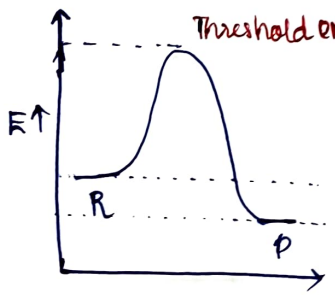
15/10/25 Wednesday.

Influence of Temperature on Reaction Rate.

* Temperature coefficient ; the ratio of rate constant of a particular rⁿ at two temperatures.

$$\frac{K_{t+10}}{K_t} \approx 2 \text{ to } 3$$

* Arrhenius Eqn → differential eqn :- $\frac{d \ln k}{dt} = \frac{E_a}{RT^2}$



Activation Energy (E_a) = Threshold energy - energy possessed by the molecule.

$$\ln K = \frac{-E_a}{RT} + C$$

$$K = A e^{-E_a/RT}$$ → Exponential form.

$$\log K = \frac{-E_a}{RT} + C$$

$$\log K \Big)_{K_1}^{K_2} = \frac{-E_a}{2.303R}\left[\frac{1}{T}\right]_{T_1}^{T_2}$$

$$\log \frac{K_2}{K_1} = \frac{E_a}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$

Q: For the reaction; $CH_{4(g)} + 2S_{2(g)} \longrightarrow CS_{2(g)} + 2H_2S_{(g)}$

$1.1\ \ell\ mol^{-1}s^{-1} \rightarrow K_1$

$K_2 \leftarrow 64\ \ell\ mol^{-1}s^{-1}$ at 823 K & 898 K :

Calculate the energy of activation.

$$\log\left[\frac{K_2}{K_1}\right] = \frac{E_a}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$

$$\log\left[\frac{64}{1.1}\right] = \frac{E_a}{2.303 \times 8.314}\left[\frac{1}{823} - \frac{1}{898}\right]$$

$$1.76 = \frac{E_a}{19.14}\left[1.0 \times 10^{-4}\right] \qquad 0.76 = \frac{E_a}{19.14} \times 10^{-4}$$

$$E_a = \frac{19.14 \times 0.76}{10^{-4}}$$

$$E_a = \frac{19.14 \times 1.76}{1 \times 10^{-4}}$$

$$= 14546.4$$

$$= 336864.$$

$$= 1.45 \times 10^{5}\ J$$

Q: The rate constant of a 1st at 25°C & 40°C are 5.7×10^{-5} $dm^3 mol^{-1} s^{-1}$ and 1.64×10^{-4} $dm^3 mol^{-1} s^{-1}$. Calculate the Ea & preexponential factor at 40°C.

$T_1 = 25°C = 273 + 25 = 298\ K$

$T_2 = 40°C = 273 + 40 = 313\ K$

$K_1 = 5.7 \times 10^{-5}$

$K_2 = 1.64 \times 10^{-4}$

$$\log\left[\frac{K_2}{K_1}\right] = \frac{Ea}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$

$$\log\left[\frac{1.64 \times 10^{-4}}{5.7 \times 10^{-5}}\right] = \frac{Ea}{2.303 \times 8.314}\left[\frac{1}{298} - \frac{1}{313}\right]$$

$$0.45 = \frac{Ea}{19.14}\left[1.60 \times 10^{-4}\right]$$

$$Ea = \frac{0.45 \times 19.14}{1.6 \times 10^{-4}}$$

$$= 5.3 \times 10^{-4}$$

$$K = A e^{-Ea/RT}$$

$$\ln K = \ln A - \frac{Ea}{RT}$$

$$\log k = \log A - \frac{Ea}{2.303\,RT}$$

$$\log A = \log k + \frac{Ea}{2.303 \times 8.314 \times 313}$$

$$= \log(1.64 \times 10^{-4}) + \frac{5.3 \times 10^{-4}}{2.303 \times 8.314 \times 313}$$

$$= -3.78 + 88.43$$

$$\log A = 84.65$$

$$A = 4.46 \times 10^{84}$$

25
nday

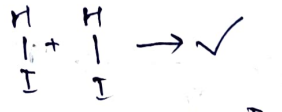
⇒ Collission Theory:

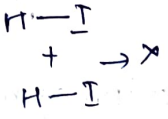
* rate of rxn is directly proportional to no. of collisions per unit volume and unit time (collision frequency)

$$\text{rate of rxn} \propto P$$

* not all collision leads to pdt formation, only effective collision do.
* for effective collision :
  * all molecules should possess threshold energy.
  * proper orientation.

$$2HI \longrightarrow H_2 + I_2$$





⇒ Enzyme Catalysis.

* Enzymes ≡ biological catalysts.
* two con-ve catalysts.

$$CH_2=CH_2 + H_2 \xrightarrow[\text{Ni}]{Pd/Pt} CH_3-CH_3$$

* Enzymes are highly efficient (≈2 fold higher than normal catalysts)

eg:

$$C_{12}H_{22}O_{11} \xrightarrow{\text{invertase}} C_6H_{12}O_6 + C_6H_{12}O_6$$

sucrose — glucose — fructose

$$C_{12}H_{22}O_{11} \xrightarrow{\text{sucrase}} 2C_6H_{12}O_6$$

glucose

$$NH_2-CO-NH_2 \xrightarrow{\text{urease}} NH_3 + CO_2$$

* enzymes are highly specific.
* The efficiency of an enzyme is max at its optimum temperature
* enzymes are also pH specific ⇒ optimum pH.
* Body fluids like blood are slightly alkaline.

⇒ Theory of Unimolecular Reactions. (only one reactant)

$$CH_3CHO_{(g)} \longrightarrow CH_4 + CO$$

* Lindemann Theory

$$A \longrightarrow \text{pdt.}$$

$$A + M \xrightarrow{K_1} A^* + M$$

$A^* + M \xrightarrow{k_2} A + M$ deactivation

$A^* \xrightarrow{k_3}$ product decomposition to pdt.

Rate of pdt $= k_3 \times [A^*]$ → since conc. of intermediate cannot be determined;

★ Steady-state approximation (SSA)

The conc. of a reactive intermediate remains constant.

ie;

Rate of formation of $[A^*]$ = Rate of decrease of $[A^*]$

$$\frac{d[A^*]}{dt} = 0$$

rate of formation of $[A^*] = k_1[A][M]$

rate of decomposition of $[A^*] = k_2[A^*][M] + k_3[A^*]$

rate of formation of A^* = rate of decomposition of A^*.

$\therefore k_1[A][M] = k_2[A^*][M] + [A^*]k_3$

$$k_1[A][M] = [A^*](k_2[M] + k_3)$$

$$[A^*] = \frac{k_1[A][M]}{k_2[M] + k_3}$$

$$\text{Rate} = \frac{k_1 k_3 [A][M]}{k_2[M] + k_3}$$

if $k_2[M] >>> k_3$; (deactivation rapid)

$$\text{Rate} = \frac{k_1 k_3 [A]}{k_2} = k'[A]$$

if $k_3 >>> k_2[M]$ (deactivatn not at all)

$$\text{Rate} = \frac{k_1 k_3}{k_2}$$

$$\text{Rate} = k_1[A][M]$$

→ Michael's Menten Kinetics. [enzyme-catalysis]

$$E + S \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} ES \quad \rightarrow \text{enzyme-substrate complex}$$

$$ES \xrightarrow{k_2} \text{pdt} + E$$

$$r = k_2[ES]$$

Applying SSA;

rate of formatn of ES = rate of decompositn of ES

$$k_1[E][S] = k_{-1}[ES] + k_2[E][S]$$

$$k_1[ES] - k_{-1}[ES] = k_2[ES]$$

$$[S][E]k_1 - k_{-1}[ES] = k_2$$

$$k_1[E][S] - k_{-1}[ES] = k_2[ES]$$

$$\frac{d(ES)}{dt} = 0, \quad k_1[E][S] = (k_{-1} + k_2)[ES] \quad -(1)$$

initial conc of enzyme = Free enzyme + ES complex.

$$[E_0] = [E] + [ES]$$

$$[E] = [E_0] - [ES] \quad -(2)$$

Substituting (2) in (1).

$$k_1 [S] \left([E_0] - [ES]\right) = (k_{-1} + k_2)[ES]$$

$$k_1 [S][E_0] - k_1 [S][ES] = [ES](k_{-1} + k_2)$$

$$k_1 [S][E_0] = [ES](k_{-1} + k_2) + k_1 [S][ES]$$

$$[ES] = \frac{k_1 [S][E_0]}{k_1 [S] + k_{-1} + k_2}$$

rate $= k_2 [ES]$

$$= k_2 \frac{k_1 [S][E_0]}{k_1 [S] + k_{-1} + k_2}$$

$$= \frac{k_2 k_1 [S][E_0]}{k_1 [S] + k_{-1} + k_2}$$

dividing by k_1

$$\text{rate} = \frac{k_2 [S][E_0]}{[S] + k_m}, \text{ where } k_m = \left(\frac{k_{-1} + k_2}{k_1}\right)$$

→ Michael's constant.

when entire Enzyme is bound into Substrate;

$$[ES] = [E_0]$$

$\therefore$

$$\text{rate} = k_2 [E_0]$$

V_{max} = The max rate happens in this condit^n.

$$\text{rate} = \frac{V_{max}[S]}{[S] + K_m}$$